



SECP2523: DATABASE

<2425/1>

DATABASE LAB ASSIGNMENT:

DDL & DML

(MODULE 1: MEMBERSHIP APPLICATION)

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MODULE 1.1: APPLY FOR MEMBERSHIP

Module Description

The main purpose of this module is to register to become a member of the Koperasi Kakitangan KADA Kelantan Berhad (KKKK). This step is essential because the user needs to apply for membership to be able to login into KKKK official website. In the first page of the application, the user must fill all the personal forms and also the family form. There are three main tables of the database to apply for membership which are the member table, membership application table and family table.

DDL Module 1.1

Create for Family Table

```
CREATE TABLE `tb_family` (  
  `f_id` int(11) NOT NULL,  
  `f_name` varchar(150) NOT NULL,  
  `f_ic` varchar(12) NOT NULL,  
  `f_relationship` varchar(50) NOT NULL,  
  `f_member_id` int(11) NOT NULL,  
  `f_created_at` timestamp NOT NULL DEFAULT current_timestamp()  
);
```

Constraint for Family Table

```
ALTER TABLE `tb_family`  
  ADD CONSTRAINT `tb_family_ibfk_1` FOREIGN KEY (`f_member_id`) REFERENCES  
  `tb_member` (`mem_id`) ON DELETE CASCADE;
```

Create for Member Table

```
CREATE TABLE `tb_member` (`mem_id` int(11) NOT NULL, `mem_ic` varchar(12) NOT NULL,
`mem_pf` int(11) NOT NULL, `mem_name` varchar(150) NOT NULL,
`mem_status` varchar(20) NOT NULL, `mem_address` varchar(100) NOT NULL,
`mem_postcode` int(5) NOT NULL, `mem_state` int(11) NOT NULL,
`mem_gender` char(1) NOT NULL, `mem_religion` varchar(20) NOT NULL,
`mem_race` varchar(20) NOT NULL, `mem_email` varchar(50) NOT NULL,
`mem_password` varchar(128) NOT NULL, `mem_position` varchar(20) NOT NULL,
`mem_position_grade` varchar(4) NOT NULL, `mem_office_address` varchar(100) NOT NULL,
`mem_office_postcode` int(5) NOT NULL, `mem_office_state` int(11) NOT NULL,
`mem_fax` varchar(11) DEFAULT NULL, `mem_tel` varchar(12) NOT NULL,
`mem_tel_house` varchar(11) DEFAULT NULL, `mem_salary` float NOT NULL,
`mem_bank` int(11) NOT NULL, `mem_bank_no` varchar(20) NOT NULL,
`mem_appdate` timestamp NOT NULL DEFAULT current_timestamp() ON UPDATE
current_timestamp(),
`mem_modal_share` float NOT NULL DEFAULT 0, `mem_modal_fee` float NOT NULL
DEFAULT 0,
`mem_fixed_saving` float NOT NULL DEFAULT 0, `mem_charity_fund` float NOT NULL
DEFAULT 0,
`mem_saving` float NOT NULL DEFAULT 0, `mem_application_id` int(11) NOT NULL
);
```

Constraint for Member Table

```
ALTER TABLE `tb_member`
ADD CONSTRAINT `tb_member_ibfk_1` FOREIGN KEY (`mem_state`) REFERENCES
`tb_state` (`state_id`),
ADD CONSTRAINT `tb_member_ibfk_2` FOREIGN KEY (`mem_office_state`) REFERENCES
`tb_state` (`state_id`),
ADD CONSTRAINT `tb_member_ibfk_3` FOREIGN KEY (`mem_bank`) REFERENCES
`tb_bank` (`bank_id`),
ADD CONSTRAINT `tb_member_ibfk_4` FOREIGN KEY (`mem_application_id`) REFERENCES
`tb_membership` (`m_app_id`);
```

Create for Membership Application Table

```
CREATE TABLE `tb_member` (`mem_id` int(11) NOT NULL, `mem_ic` varchar(12) NOT NULL,
`mem_pf` int(11) NOT NULL, `mem_name` varchar(150) NOT NULL,
`mem_status` varchar(20) NOT NULL, `mem_address` varchar(100) NOT NULL,
`mem_postcode` int(5) NOT NULL, `mem_state` int(11) NOT NULL,
`mem_gender` char(1) NOT NULL, `mem_religion` varchar(20) NOT NULL,
`mem_race` varchar(20) NOT NULL, `mem_email` varchar(50) NOT NULL,
`mem_password` varchar(128) NOT NULL, `mem_position` varchar(20) NOT NULL,
`mem_position_grade` varchar(4) NOT NULL, `mem_office_address` varchar(100) NOT NULL,
`mem_office_postcode` int(5) NOT NULL, `mem_office_state` int(11) NOT NULL,
`mem_fax` varchar(11) DEFAULT NULL, `mem_tel` varchar(12) NOT NULL,
`mem_tel_house` varchar(11) DEFAULT NULL, `mem_salary` float NOT NULL,
`mem_bank` int(11) NOT NULL, `mem_bank_no` varchar(20) NOT NULL,
`mem_appdate` timestamp NOT NULL DEFAULT current_timestamp() ON UPDATE
current_timestamp(),
```

Constraint for Membership Application Table

```
ALTER TABLE `tb_membership`
ADD CONSTRAINT `tb_membership_ibfk_1` FOREIGN KEY (`m_bank`) REFERENCES
`tb_bank` (`bank_id`),
ADD CONSTRAINT `tb_membership_ibfk_2` FOREIGN KEY (`m_bod_id`) REFERENCES
`tb_bod` (`b_id`),
ADD CONSTRAINT `tb_membership_ibfk_3` FOREIGN KEY (`m_office_state`) REFERENCES
`tb_state` (`state_id`),
ADD CONSTRAINT `tb_membership_ibfk_4` FOREIGN KEY (`m_state`) REFERENCES
`tb_state` (`state_id`),
ADD CONSTRAINT `tb_membership_ibfk_5` FOREIGN KEY (`m_appstatus`) REFERENCES
`tb_appstatus` (`app_id`);
```

DML Module 1.1

Insert data into Family Table

```
INSERT INTO tb_family (f_name, f_ic, f_relationship, f_member_id)
VALUES ('$name', '$ic', '$relationship', '$fnostaff')
```

Insert data into Member Table

```
INSERT INTO tb_member (
    mem_id, mem_ic, mem_pf, mem_name, mem_status, mem_address, mem_postcode,
    mem_state, mem_gender, mem_religion,
    mem_race, mem_email, mem_password, mem_position, mem_position_grade,
    mem_office_address, mem_office_postcode, mem_office_state, mem_fax,
    mem_tel, mem_tel_house, mem_salary, mem_bank, mem_bank_no, mem_modal_share,
    mem_fixed_saving, mem_charity_fund, mem_saving, mem_application_id
) VALUES (
    '$fnostaff', '$fic', '$fnopf', '$fname', '$fstatus', '$fhaddr', '$fhpostcode', '$fhstate', '$fgender',
    '$freligion',
    '$frace', '$femail', '$hashedPassword', '$fposi', '$fgrade', '$foaddr', '$fopostcode', '$fostate',
    '$fnofax',
    '$fnotel', '$fnohouse', '$fsalary', '$fbank', '$fnoacc', 0.00, 0.00, 0.00, 0.00, '$sapp_id')
```

Insert data into Membership Application Table

```
INSERT INTO tb_membership (
    m_nostaff, m_ic, m_pf, m_name, m_status, m_address, m_postcode, m_state, m_gender,
    m_religion,
    m_race, m_email, m_password, m_position, m_position_grade, m_office_address,
    m_office_postcode, m_office_state, m_fax,
    m_tel, m_tel_house, m_salary, m_bank, m_bank_no
) VALUES (
    '$fnostaff', '$fic', '$fnopf', '$fname', '$fstatus', '$fhaddr', '$fhpostcode', '$fhstate', '$fgender',
    '$freligion',
    '$frace', '$femail', '$hashedPassword', '$fposi', '$fgrade', '$foaddr', '$fopostcode', '$fostate',
    '$fnofax',
    '$fnotel', '$fnohouse', '$fsalary', '$fbank', '$fnoacc'
)
```

MODULE 1.2: APPROVE MEMBERSHIP APPLICATION

Module Description

After the user successfully submit their member application the clerk will either approve or reject the application form. For approval, the applicant is added to the system as a user with their credentials stored in the database, and their application status is updated to "approved", then the user officially becomes a member and has the authentication to login to the website. For rejection, their application status is marked as "rejected," and their record is deleted. The user receives feedback through alert messages indicating whether the applicant has been approved or rejected.

DDL & DML Module 1.2

Create User Table

```
CREATE TABLE `tb_user` (  
  `u_id` int(11) NOT NULL,  
  `u_password` varchar(128) NOT NULL,  
  `u_role` int(11) NOT NULL  
)
```

Create Password form Membership Application Table

```
SELECT m_password FROM tb_membership  
WHERE m_app_id = $membership_id
```

Update the Status of the applicant into approve/reject

```
UPDATE tb_membership  
SET m_appstatus = 2 , m_bod_id = $uid  
WHERE m_app_id = $membership_id"
```

```
UPDATE tb_membership  
SET m_appstatus = 3 , m_bod_id = $uid  
WHERE m_nostaff = $nostaff
```

Insert authentication ID after approve

```
INSERT INTO tb_user (  
  u_id, u_password, u_role  
) VALUES (  
  '$nostaff', '$pass', 1  
)
```

Delete the information after reject

```
DELETE FROM tb_member  
WHERE mem_id = $nostaff
```